

Analysis of Historical Photographs in The Daily Mirror Newspaper

YOLO-based Image and Caption Detection

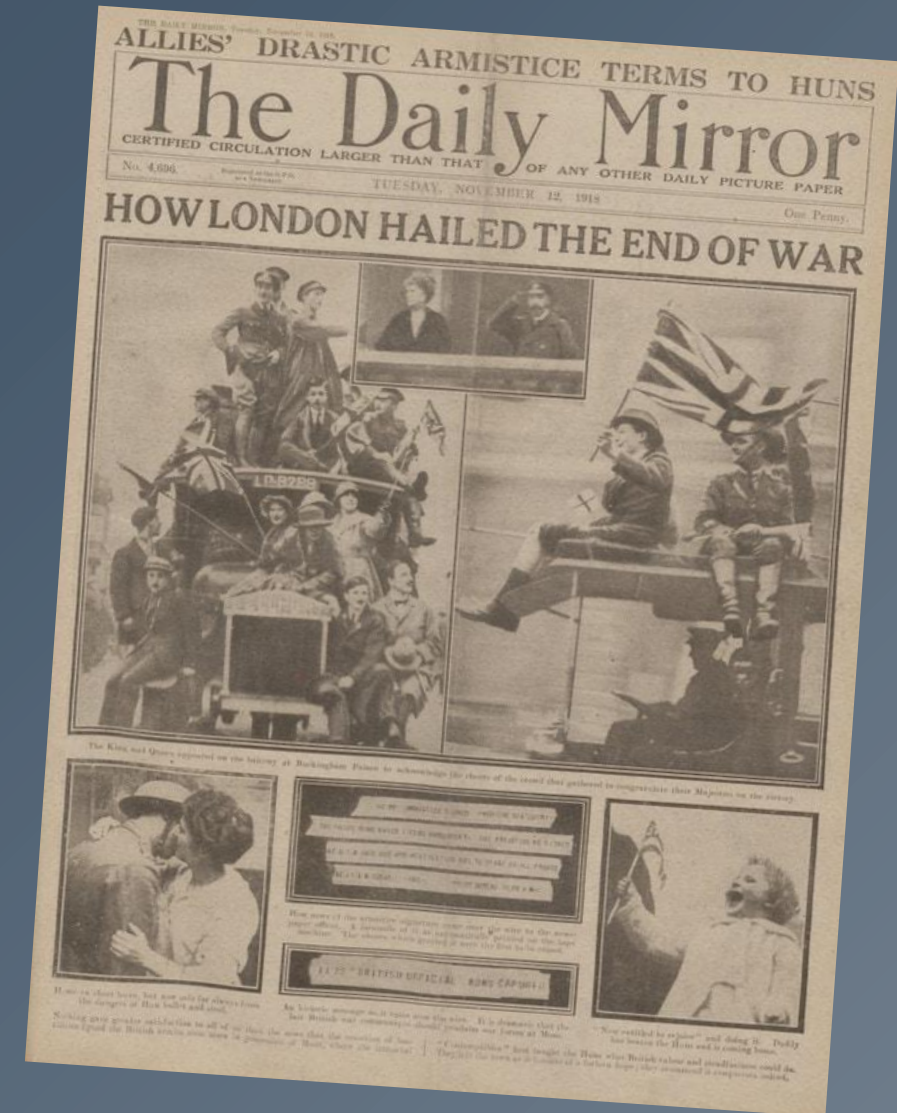
Supervisor: Isabelle Bloch

Support: Daniel Foliard, Mohamed Chelali – Historian Team at Université Paris Cité

Nesrine Naaman

Sorbonne University

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Contents

01

Introduction

02

Bibliography and Pipeline

03

Methodology: YOLOv8

04

Results and Issues

05

Post-Processing

06

Summary and Next Steps

INTRODUCTION

Motivation: Turn Newspapers from Static Scans into Living Queryable Historical Data

- ✧ Images carry historical meaning that text alone misses
- ✧ Images without their captions can be ambiguous

→ We want to extract newspaper photos and captions and make use of both and their connection

★ NEWSPAPER ⇒ The Daily Mirror

Aim: Develop an **Automated Pipeline** for Detecting, and Annotating the Daily Mirror images and captions → enable the Structured Digitization and Semantic Understanding of the complex Newspaper layouts

BIBLIOGRAPHY AND PIPELINE

⚡ Pipeline:

1 Visual Content Recognition

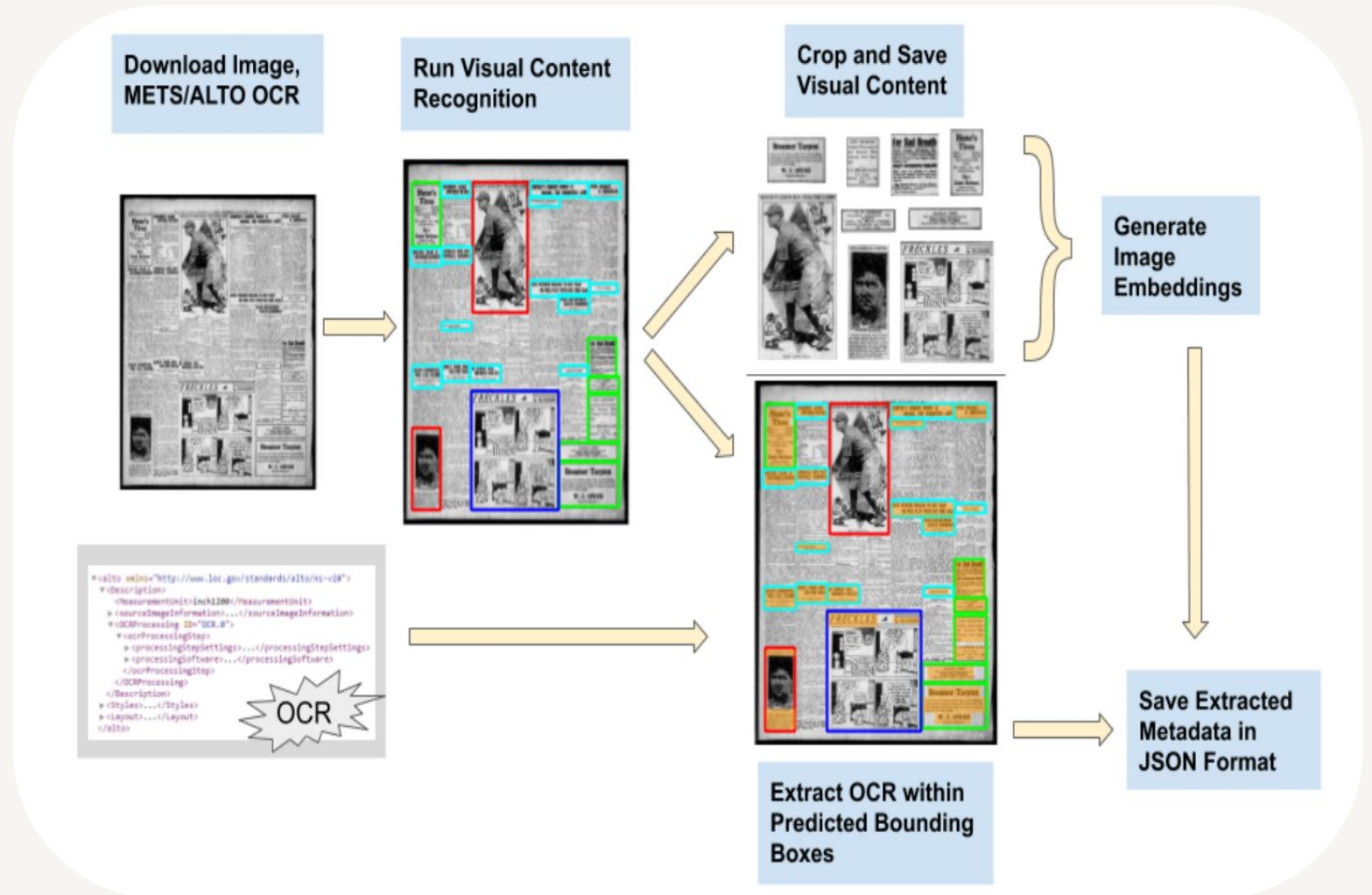
Object Detection to localize Images and Captions within Bounding Boxes using YOLO

2 OCR on Detected Captions

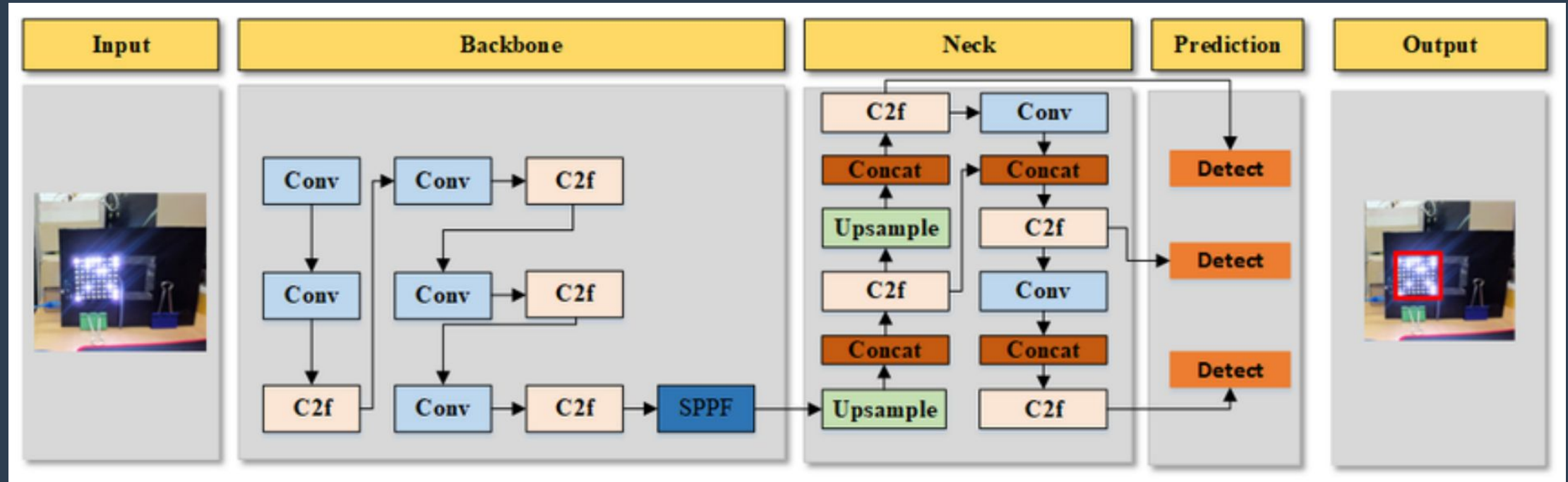
Optical Character Recognition (OCR) systems like Tesseract OCR are applied to the detected text regions to extract machine-readable textual content

3 Projection into multimodal embedding space

The visual features and textual embeddings are projected into a shared multimodal embedding space to verify whether a caption correctly describes its associated image



Methodology: YOLOv8 - You Only Look Once



YOLOv8 : Real-Time Detection

01 Backbone

Feature Extraction Network: Deep convolutional neural network that extracts hierarchical features from images

- ✓ Low-level features (edges, textures)
- ✓ Mid-level features (shapes, patterns)
- ✓ High-level features (objects, semantics)

02 Neck

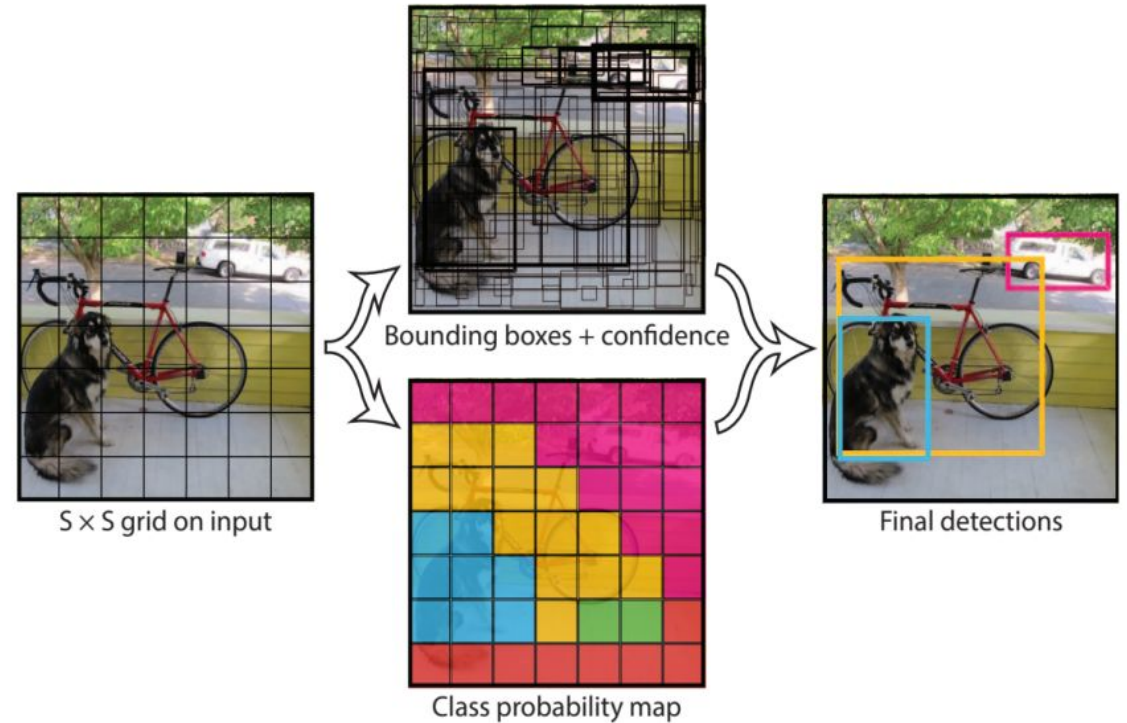
Feature Pyramid Network: Aggregates features from different backbone layers to enhance multi-scale detection capabilities

03 Head

Detection Layer: Head outputs for each cell BB coordinates, class probabilities, and confidence scores for each BB

YOLOv8 : Post-Processing

Non-Maximum Suppression (NMS) :
multiple overlapping boxes for the same
object $\rightarrow$ keep only the box with the highest
confidence and discard others with high IoU
overlap



✧ 1000 Daily Mirror
Newspaper Pages



Training & Datasets

✧ 197 Pages annotated

✧ JSON file of annotation
follows COCO format

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    {
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      "arkindex_id": "4fe2653f-6ea9-4e07-846b-c4878eea21fc",
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      "height": 2096
    },
    {
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      "width": 1653,
      "height": 2089
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      "arkindex_id": "a02729f6-f7a8-4fa1-9568-678b893f605a",
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      439.0
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        865,
        472,
        1304,
        1184,
        1304,
        1184,
        865
      ]
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    "image_id": 1,
    "category_id": 2,
    "bbox": [
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      1309.0,
      725.0,
      34.0
    ],
    "segmentation": [
      [

```


Training & Datasets

✧ Modified format to yolo_labels
needed by yolov8n model

✧ Split annotated data:
80% train 20% test

✧ Used yolov8n.pt model **pre-trained**
on COCO data and fully **fine-tuned** it on
our training set

```
2 0.49788263762855417 0.196073679108095 0.9328493647912885 0.03926320891904993
1 0.2616454930429522 0.9471643238002908 0.3720508166969147 0.02035870092098885
3 0.7253478523895946 0.7944740668928745 0.4476709013914096 0.28211342704798836
3 0.8239564428312159 0.39820649539505576 0.23472474289171205 0.35239941832283084
1 0.8266787658802178 0.5855550169655841 0.2026618269812462 0.016480853126514785
1 0.38324258923169996 0.586282113427048 0.4869933454325469 0.012118274357731459
1 0.49939503932244406 0.9755210857973825 0.9249848759830611 0.04023267086766844
3 0.27404718693284935 0.7976248182258846 0.4476709013914096 0.28259815802229765
3 0.37840290381125224 0.39650993698497333 0.647912885662432 0.35579253514299564
2 0.4903206291591047 0.6349975763451284 0.8802177858439202 0.036839554047503636
1 0.7265577737447066 0.9471643238002908 0.3424077434966727 0.02035870092098885
1 0.5018148820326679 0.6015511391177897 0.9044162129461585 0.031022782355792537
```

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<class_id>:

{ 0: 'Subcaption', 1: 'Caption', 2: 'Title', 3: 'Image', 4: 'Subimage' }

Train set



Test set

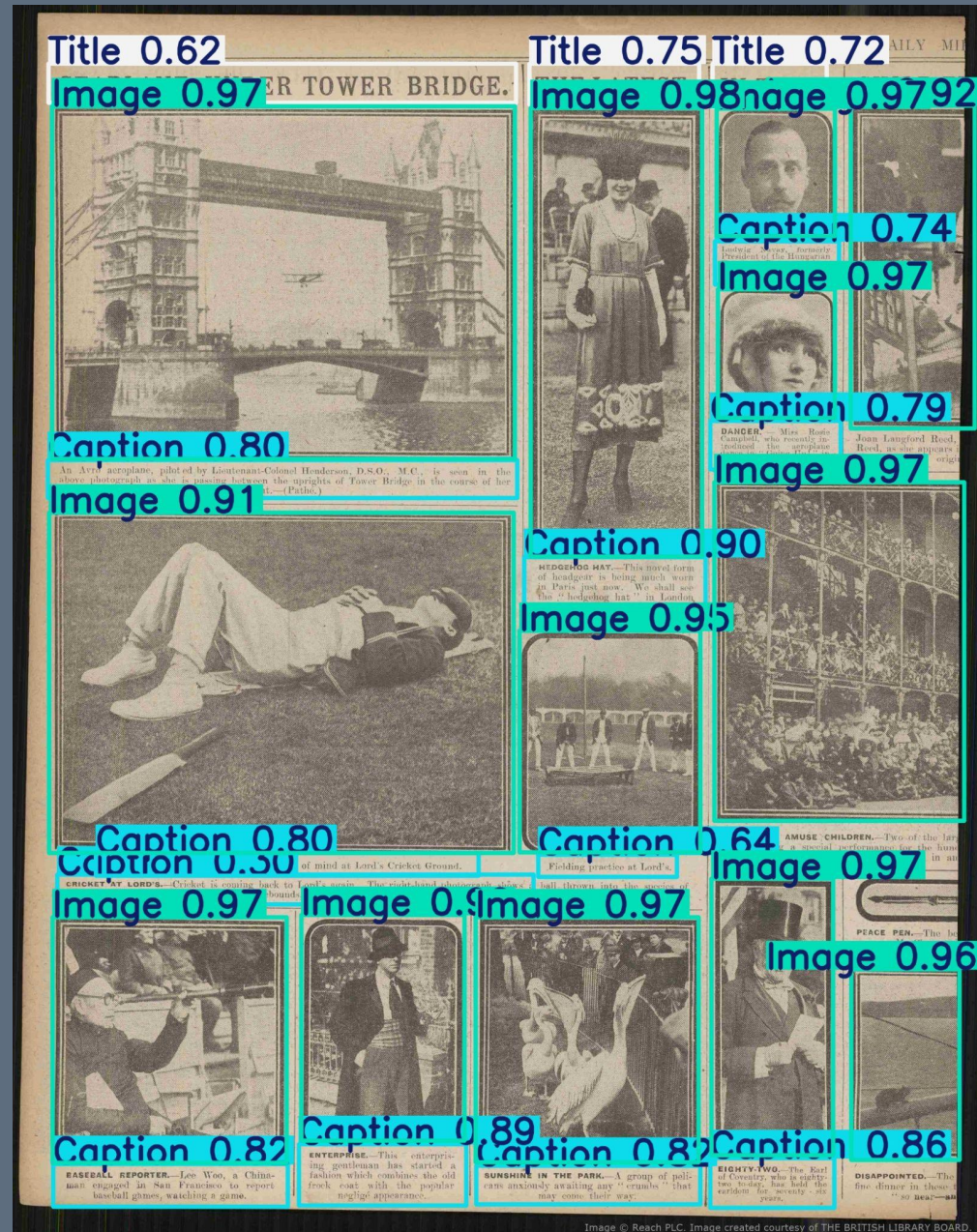


Model Results on the:

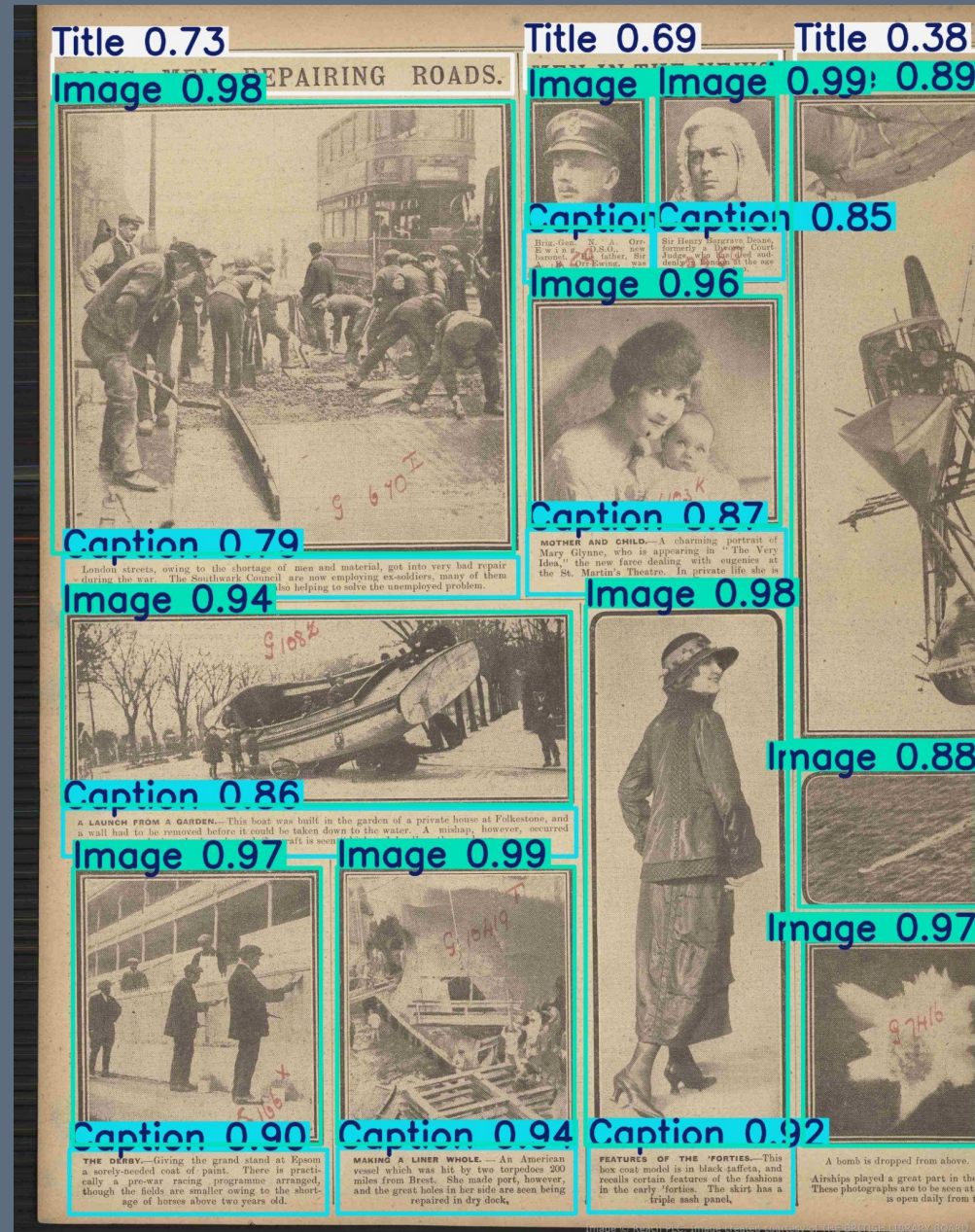
Unannotated set



Model Results on the: Unannotated set



Model Results on the: Unannotated set



Issues

✦ Rotated Images are NOT detected ✦

'ACTED LIKE A CHARM'



Eczema Permanently Banished by Zam-Buk

"MY daughter, Olive May, suffered dreadfully from eczema," says Mrs. Matthews, 11, Bertal Road, Tooting, London, S.W. "Unsightly places broke out on Olive's leg, and also behind her ear and on her chin. They soaked the child miserable.

After trying in vain to heal Olive's sores with lotions and ordinary ointment, I took her in succession to three doctors, but their preparations also quite failed to cure the eczema,

Caption 0.54

Olive May Matthews, of Tooting.

✦ Too many Caption BBs for some examples ✦



Caption 0.84
Caption 0.85
Caption 0.73

London was in deep mourning yesterday for Nurse Cavell, whose remains were borne through London on the way to Norwich—her native city—for interment in the cathedral, and thousands of people assembled along the route to pay their last tribute to the woman who unflinchingly faced the rifles of a German firing squad. The British will never forget the Cavell and Fryatt crime, committed by the enemy for sheer hate of us.—(Daily Mirror photographs.)

Earl of Athlone greets Queen Alexandra.

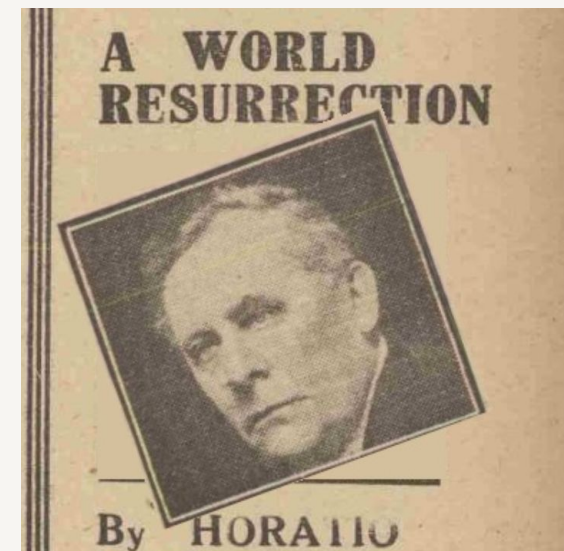
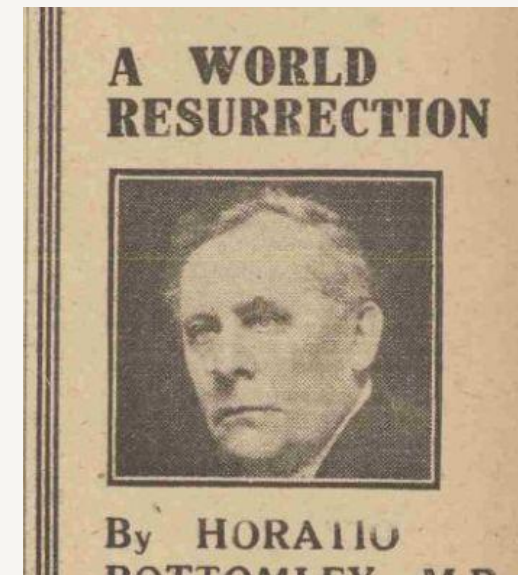
Rotation Issue Solution

Data Augmentation

We **Augmented** our training and test data by manually rotating the last image of every newspaper page to one of the following angles randomly: [-15, -10, -5, +5, +10, +15]

The background color is the color of the closest pixel right outside the BB

We fine-tuned a new pre-trained [yolov8n.pt](https://github.com/ultralytics/yolov8) model with all our augmented dataset



Rotation Issue Solution

Some Examples

The respective annotation is updated: The new BB of the rotated image includes a big Box encapsulating all 4 corners of the rotated image and some of the background

ad Saturday. advertising "ten-day ne" for £5 15s. 6d., ad to the Italian lakes

POPULAR. fares, as the ad- old me, included the Wales 17s., Carlisle Non-stop corridor ex- return journey from for 3s. 3d.

more noticeable in the which I found in the

aper there is an article ad prices from which on is very dear this a pound. Hake has popular, sending the to the Easter Monday the week-end doings

ere is a picture of fly- ing an archaic-looking e in which I doubt if ld care to risk his life. as a holiday attraction and Hendon. At the ing and 'looping' by tors," also "a tyelvo- ere announced.

a day were Suffragette n smashing things at -Mr. Bernard Shaw's ion," in which the ar-word occurred—the controversy—and ars were to be cheaper!

ADVERTISED. world was not entirely

later, on the Wednes- day is splashed across is getting herself into States.

had been ordered to l to Tampico, and it l to read in a head- see," says Mr. Wilson,

pay a tax rather than marry a woman for whom he has neither respect nor affection.

Everyone has some selfish reason for getting married; it may be disguised under the gossamer veil of love; it may be, frankly, for a home or for the wherewithal to keep up a home; for a sick nurse or a housekeeper.

A still more ridiculous proposal, however, is to tax the spinster.

The vision of a husband, home and children is enshrined in the hearts of the majority of

Tony Weller's advice to his son, and this warning should be blazoned forth from every hoarding in our public streets.

The "Merry Widow" is a danger to mankind, and she is making the lot of the spinster infinitely harder.

She should be forbidden by law to remarry, or she should be taxed if she so far forgets what is due to her dead husband by taking another partner of her joys and sorrows.

The matter is an urgent one, and should be dealt with at once. Tax the "Merry Widow"!

While the machine answers will be given. If you want to "doos?" ask the pilot to know or air an

TREAT THE N

Leather clothing aircraft firm, and it warm the weather s essential.

In summer one fi centigrade at two o only the other day centigrade was regi

When climbing i every care, for the or an aeroplane.

If the passengers' when climbing in, fabric of the lower an aluminium plate use as a step.

Find out exactly i in, otherwise some d chine might detach

Once in the cockp and fasten the safe portant item. If a chine don't neglect belt, the passenger pilot will not perfor

When the engine the body rigid as t a shock; the body there is nothing to

Make sure that gloves are perfectly habit of coming ad

When all is ready the checks" (triang the wheels to keep t machine will rush a

WHEN THE EART

Don't look straig leaves the ground; earth sinking below d been ordered to o Tampico, and it to read in a head- says Mr. Wilson, ed."

L crowds in Easter nt." More than e leave London bo- Saturday.

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by imposing a tax upon him—though how this will help the issue at stake one fails to see.

Any man worthy of the name would gladly pay a tax rather than marry a woman for whom he has neither respect nor affection.

Everyone has some selfish reason for getting married; it may be disguised under the gossamer veil of love; it may be, frankly, for a home or for the wherewithal to keep up a home; for a sick nurse or a housekeeper.

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what is due to my dead husband by taking another partner of her joys and sorrows.

The matter is an urgent one, and should be dealt with at once. Tax the "Merry Widow"!

think they cannot do better than step into his shoes.

"Beware of the widders" was the immortal Tony Weller's advice to his son, and this warning should be blazoned forth from every hoarding in our public streets.

The "Merry Widow" is a danger to mankind, and she is making the lot of the spinster infinitely harder.

She should be forbidden by law to remarry, or she should be taxed if she so far forgets what is due to my dead husband by taking another partner of her joys and sorrows.

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SOMEWHERE IN RUSSIA.—A party of British bluejackets after being landed. They are fully armed.

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SHALL WE HAVE A FAMINE IN ACTORS?

FILM PLAYING MORE ATTRACTIVE THAN THE STAGE.

By RICHARD GEORGE.

BECAUSE of the discomforts of the life chorus girls are refusing to go on tour.

SHALL WE HAVE A FAMINE IN ACTORS?

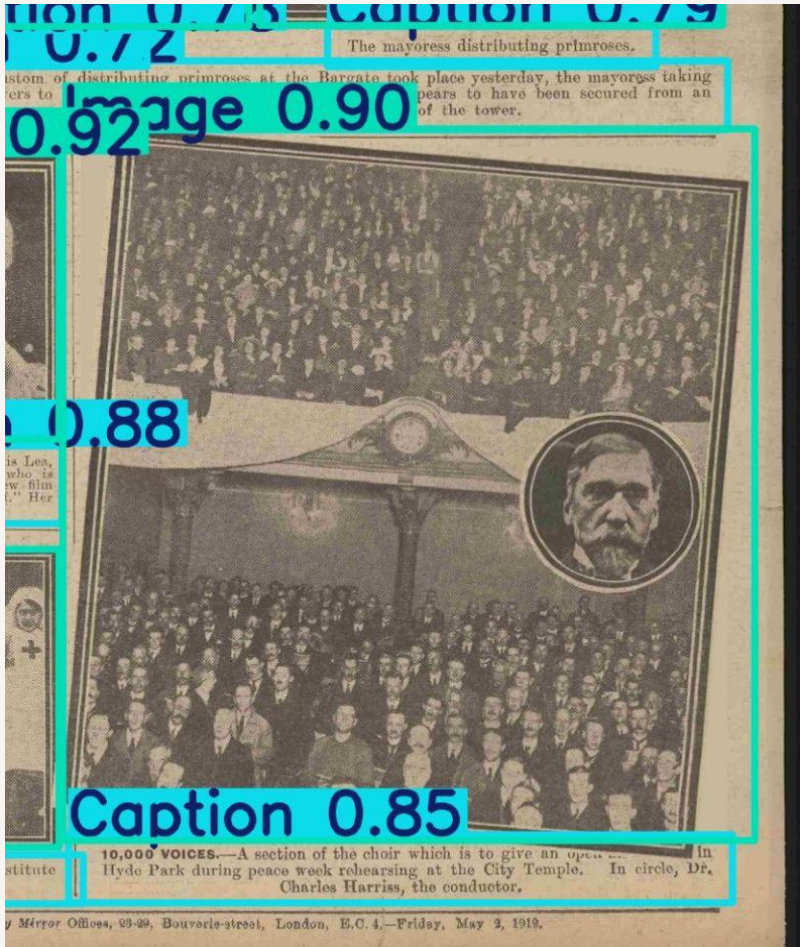
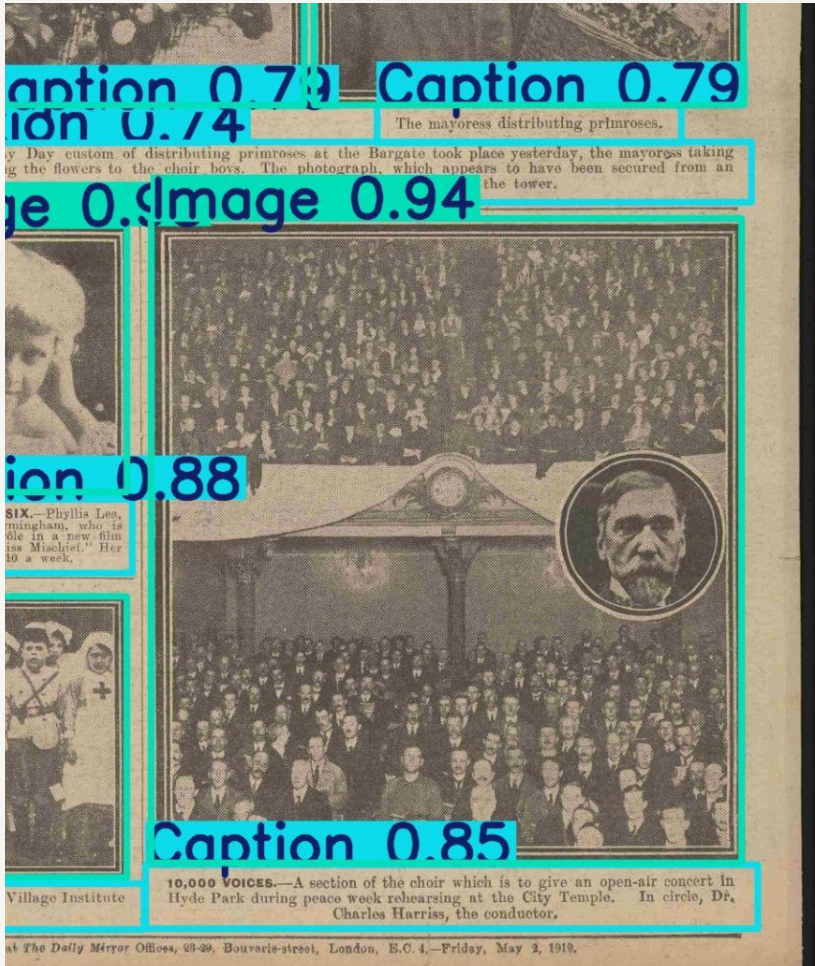
FILM PLAYING MORE ATTRACTIVE THAN THE STAGE.

By RICHARD GEORGE.

BECAUSE of the discomforts of the life chorus girls are refusing to go on tour, and I am now told there are not enough

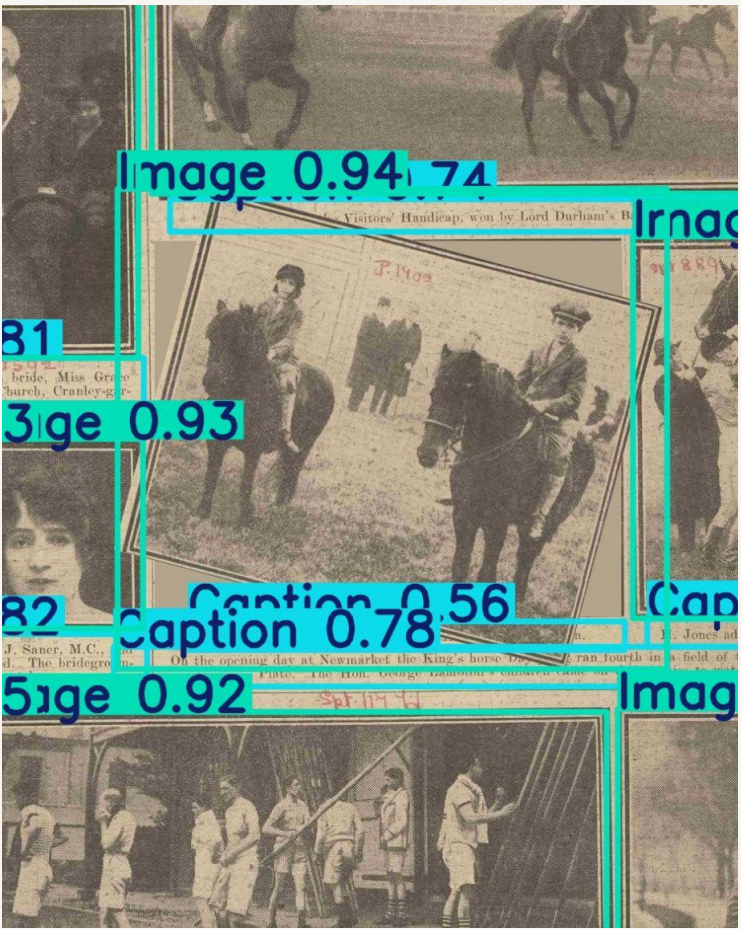
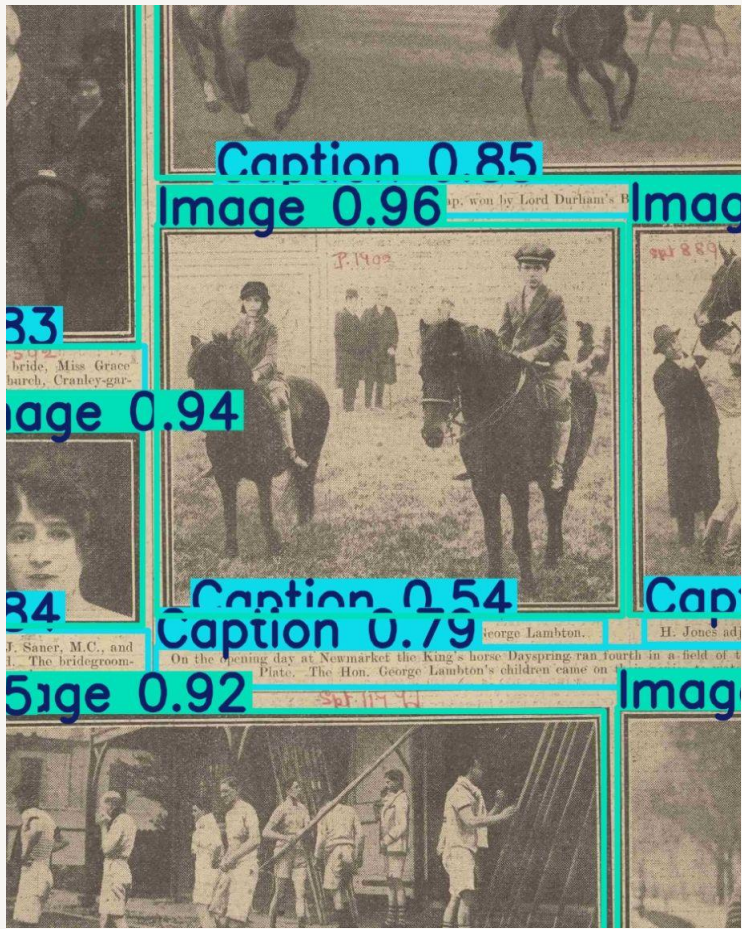
Rotation Issue Solution

Some Examples



Rotation Issue Solution

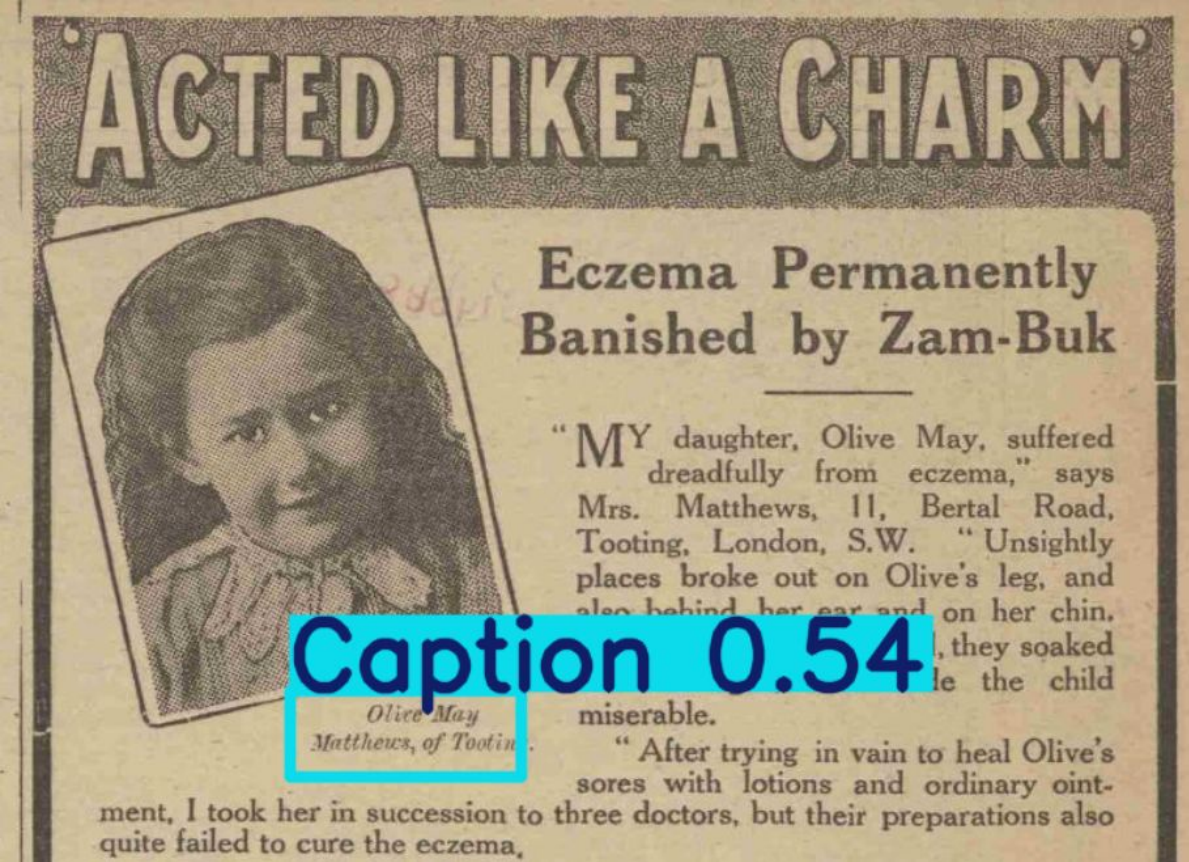
Some Examples



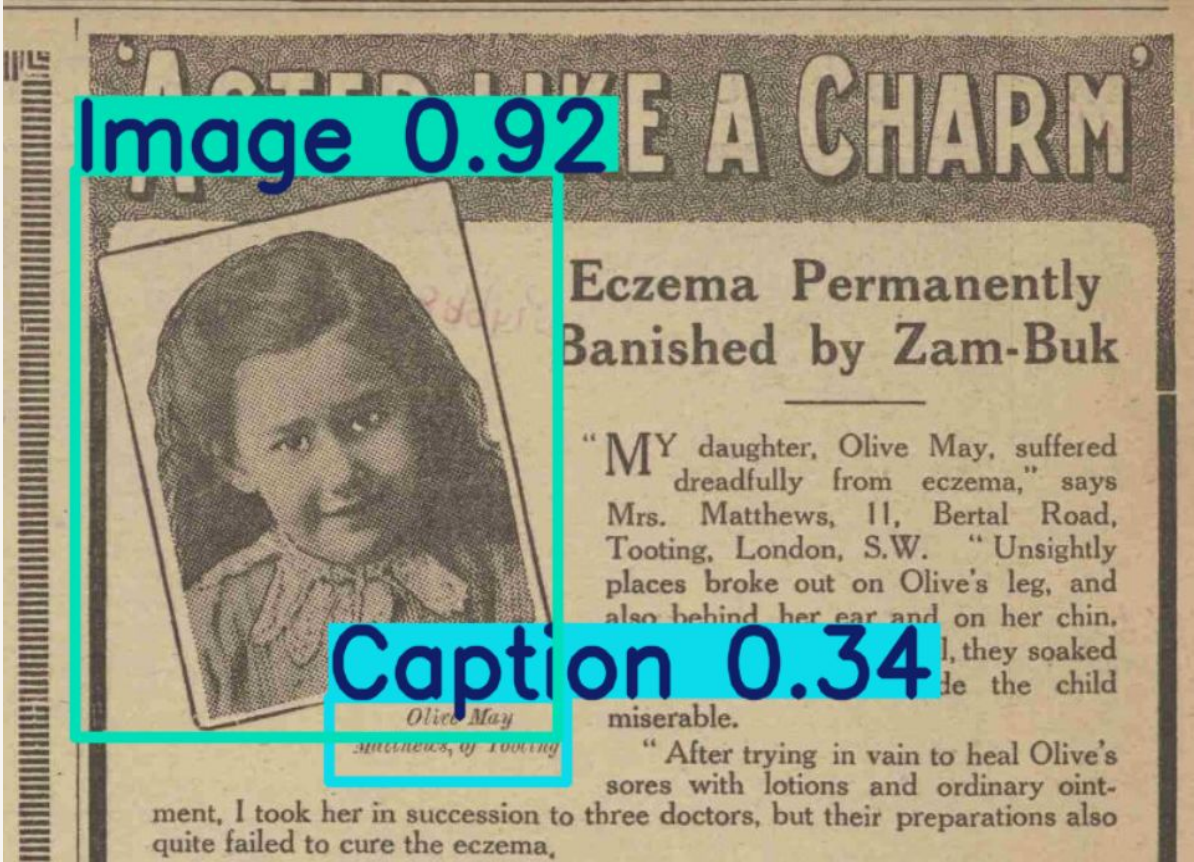
Rotation Issue Solution

Issue Solved

Before Data Augmentation



After Data Augmentation



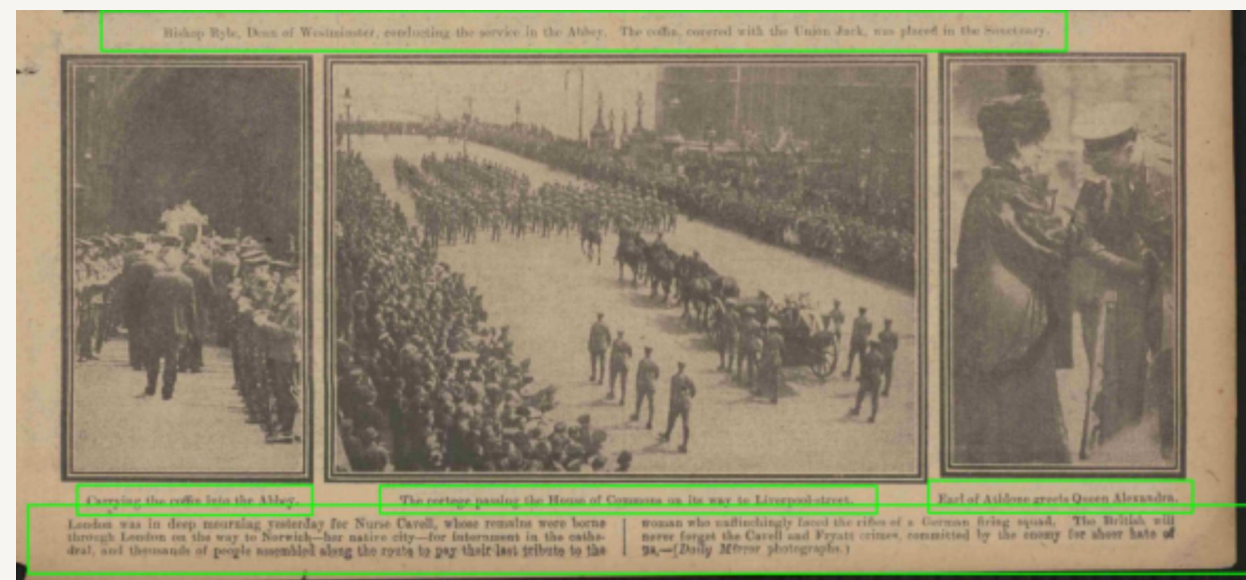
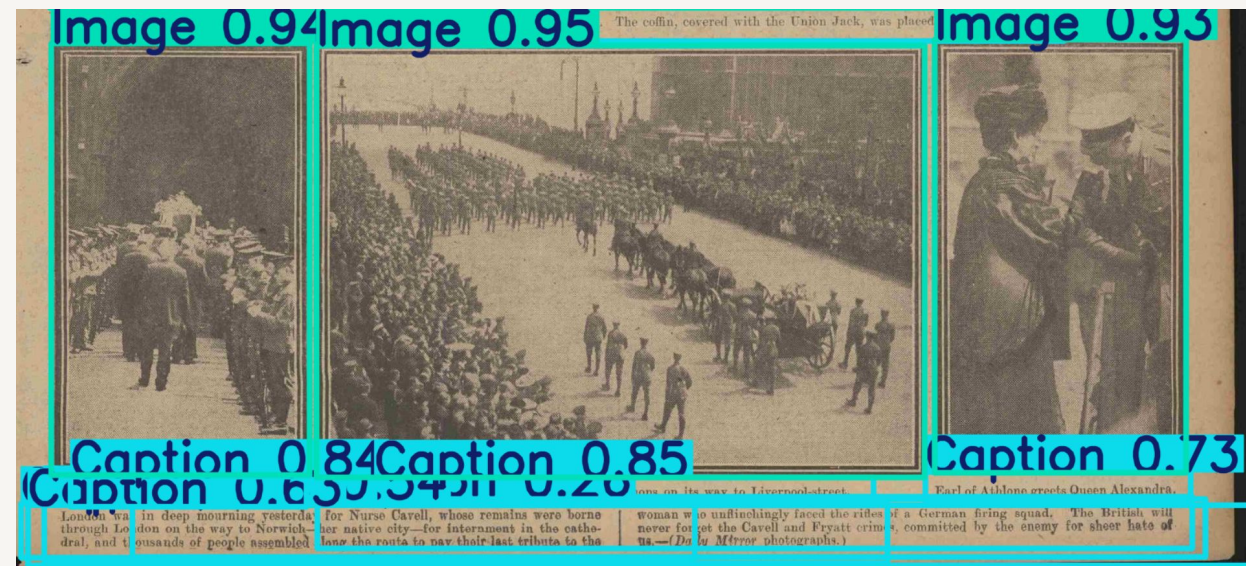
Caption BBs Issue Solution

Merging of Overlapping BBs

We merged Bounding Boxes that had an IOU greater than 0.1 (more than 10% overlap)

We modified the annotations accordingly

Goal: Make it easier and more reliable for OCR later when it uses the caption BBs



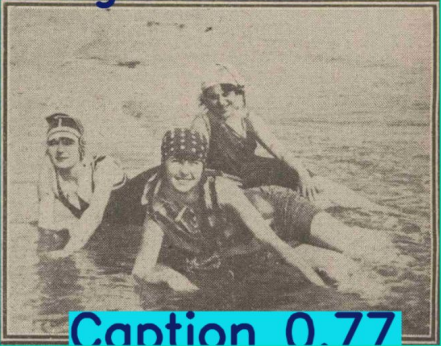
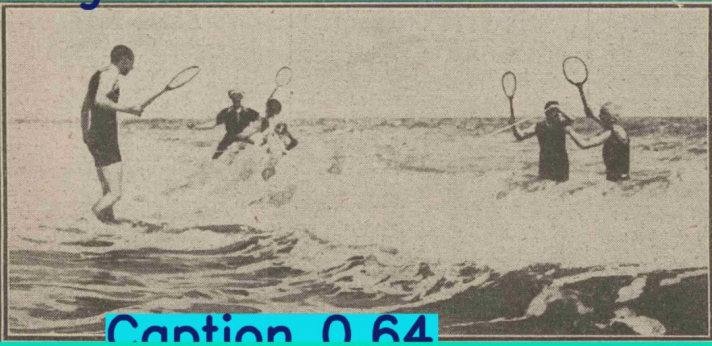
Caption BBs Issue Solution

Image 0.92

CERTAIN WAYS OF KEEPING COOL IN

Image 0.92

OF SUMMER.



Caption 0.64

Caption 0.47

Caption 0.37

Caption 0.77

Summer is here—at last—and all those who are fortunate enough to be able to do so are spring from town to the breezes of the seaside. It is difficult to keep cool anywhere these days, but, as the above photographs show, the task is not impossible, providing a certain amount of ingenuity is exercised.

One way of keeping cool—a novel tennis match in progress at Alibonagh.

Three fair Welsh maidens surf bathing.

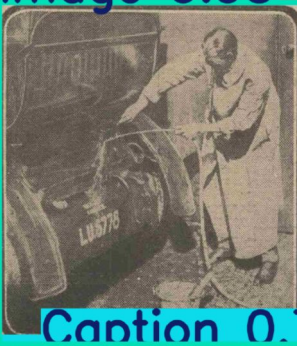
Image 0.94

ER ROLE: FA

Image 0.95

Image 0.96

R BUSINESS



Caption 0.74

Caption 0.79

Caption 0.85

Caption 0.75

Caption 0.50

Mr. Basil Foster, who is a demobilised lieutenant-colonel, has opened a garage at 2, Dorsetport-mews, Hyde Park, W.2, and will drive anyone anywhere at a moment's notice. He has at present four cars, with power to add to the number, and, in addition to driving himself, employs 1914 Service men.—(Daily Mirror photographs.)

He works hard. Not only does he paint the cars—

—but he washes them so well.

Entering up the day's orders with an employee.

THE MAY HEAT WAVE: CERTAIN WAYS OF KEEPING COOL IN THE SWELTERING HEATS OF SUMMER.

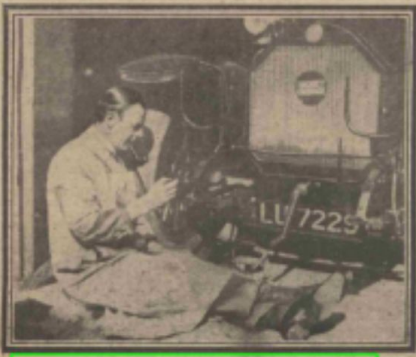


One way of keeping cool—a novel tennis match in progress at Alibonagh.

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Summer is here—at last—and all those who are fortunate enough to be able to do so are spring from town to the breezes of the seaside. It is difficult to keep cool anywhere these days, but, as the above photographs show, the task is not impossible, providing a certain amount of ingenuity is exercised.

BASIL FOSTER IN ANOTHER ROLE: FAMOUS BAT AND RACQUETS CHAMPION STARTS A MOTOR BUSINESS



He works hard. Not only does he paint the cars—

—but he washes them so well.

Entering up the day's orders with an employee.

Mr. Basil Foster, who is a demobilised lieutenant-colonel, has opened a garage at 2, Dorsetport-mews, Hyde Park, W.2, and will drive anyone anywhere at a moment's notice. He has at present four cars, with power to add to the number, and, in addition to driving himself, employs 1914 Service men.—(Daily Mirror photographs.)

Download Image,
METS/ALTO OCR



Run Visual Content
Recognition



Crop and Save
Visual Content



Generate
Image
Embeddings



Save Extracted
Metadata in
JSON Format

Extract OCR within
Predicted Bounding
Boxes

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</alto>
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OCR

Summary

Running YOLOv8 on The Daily Mirror newspaper to detect images and captions BBs + confidence score

Getting Annotations from the model and writing them in a JSON file similar to COCO format

Augmenting dataset with manually rotated images and retraining model to detect rotated images

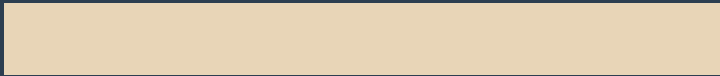
Merging captions BBs that have more than 10% overlap

Next Steps

Running OCR on caption annotations

Projecting visual features and textual embeddings into a shared multimodal embedding space

Using multimodal models like OpenCLIP to align visual and textual modalities for image-caption associations



**A Special Thank You to the Historian Team of
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annotations:**

Dr. Mohamed Chelali and Dr. Daniel Foliard

References

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- <https://dl.acm.org/doi/epdf/10.1145/3340531.3412767>
- <https://www.researchgate.net/publication/383494295> What is YOLOv8 An In-Depth Exploration of the Internal Features of the Next-Generation Object Detector
- <https://www.researchgate.net/publication/311609522> You Only Look Once Unified Real-Time Object Detection